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EURONEXT

DLT for Bond Markets: A Sustainable and Scalable Infrastructure for Euronext

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Course: Finance LAB

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Bond Market overview

Milan → largest Bond trading venue
with ≃ 270 B€

Trading Year	Market	Trading Days	Trade Volumes [m€]
2023	Amsterdam	255	3.215
2023	Brussels	255	115
2023	Dublin	254	0
2023	Lisbon	255	143
2023	Milan	254	269.886
2023	Oslo	251	48.358
2023	Paris	255	1.012

Italian Bond Market overview

MOT

Launched in 1994. Provides access for ***both professional and retail investors.***

- The only Italian **regulated market**
- European **turnover and trades leader** .
- **Order-driven**[1] **and automated**

2 segments that trade ABS, supranational, corporate etc.

1. Domestic MOT
2. EuroMOT

EURONEXT ACCESS MILAN

Launched in 2009. Dedicated to ***professional investors only.***

- **Not regulated** (MTF) under EU Directive
- **Fewer participants** and **lower volumes**
- **Order-driven**

Platform for ABS, commercial papers, infra bond , SMEs and debt instruments with participation features.

EUROTLX Bond-X

Acquired in 2013. Dedicated to **professional investors only**.

- Non regulated market (MTF)
- Requires **market makers and specialists for listing**.
- Minimum book time for specialists and market makers

Dedicated to bank and corporate Eurobonds

Source: Euronext Factbook 2023; Euronext.com

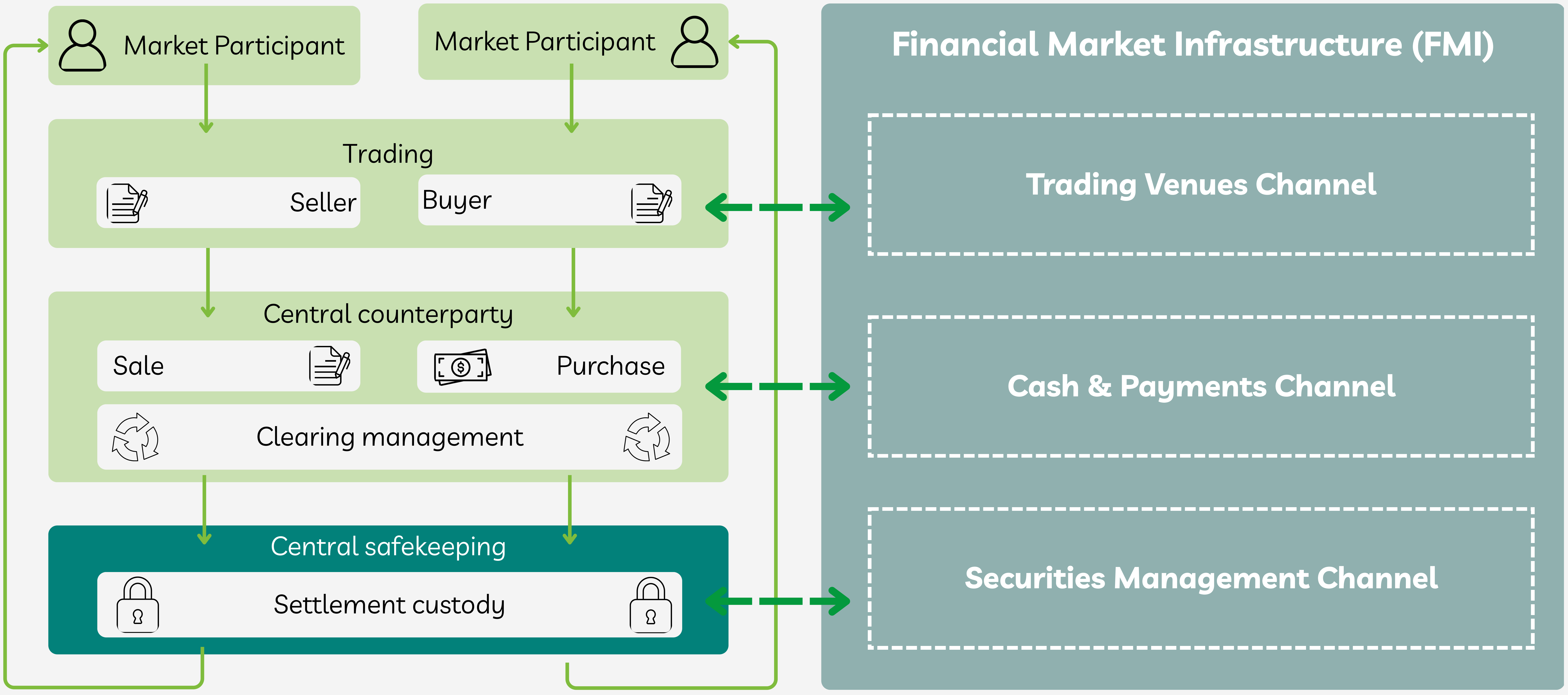
(1) Order Driven Market: atrading system where buy and sell orders from investors are matched based on predefined rules, primarily price and time, without intermediaries setting prices.

Current Euronext Infrastructure

Euronext offers a wide variety of services ⇒ highly sophisticated internal technical infrastructure.

1. **Trade and Distribution Services (Market Venues)**
2. **Central Securities Depository (CSD)**
3. **Clearing and Settlement Services**

Bond Trading and FMI Structure



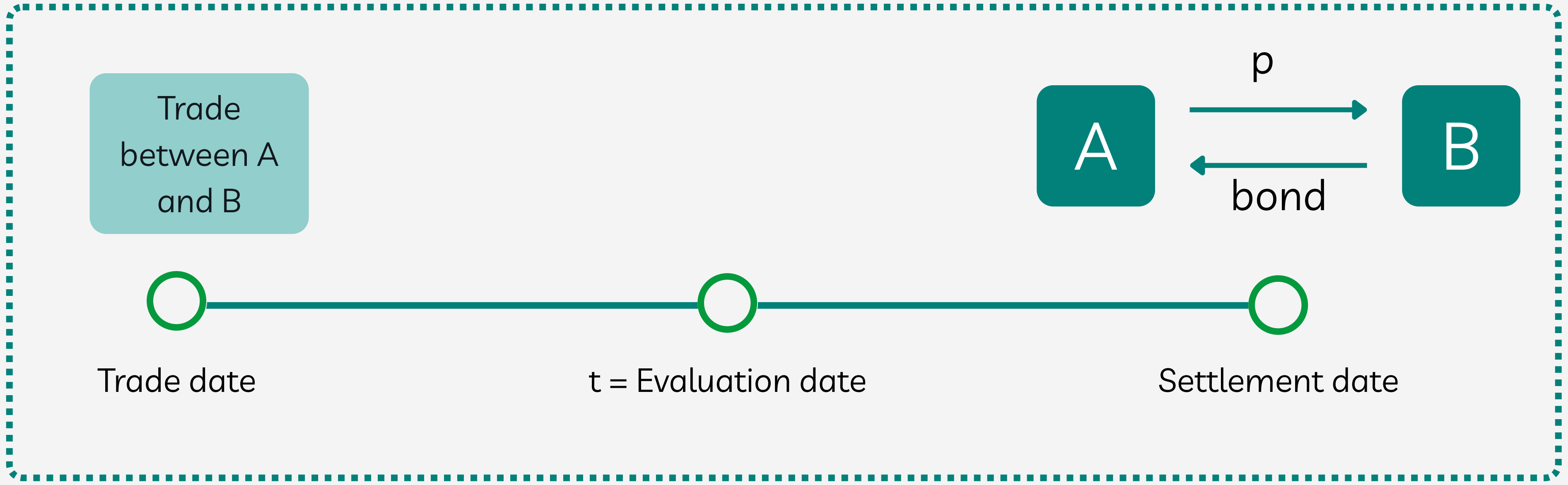
C&S Services Rationale

In order to face counterparty default risk, liquidity risks in the midst of a trade, a large pool of collateral is necessary. That being, a Clearing & Settlement infrastructure becomes fundamental.

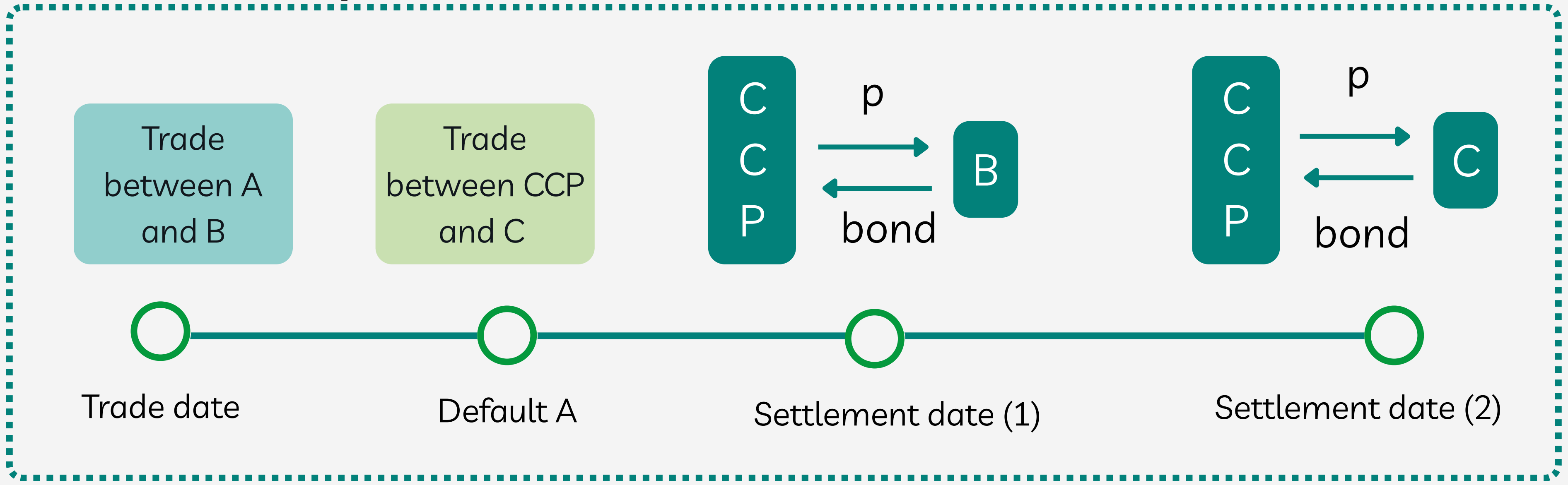
Current FMI

- T+2
- Multiple CSDs
- Multiple banking infra
- Complex compliance systems

Cash trade. No default.



Close out of cash position — Default of A & CCP intervention



Source: Euronext Clearing Methodologies. VaR Margins

2 Technological Background

2.1 DLT and Blockchain

2.2 Interoperability

2.3 Case study: Cassa Depositi e Prestiti

2.4 Digital currencies: CBDCs and Stablecoins

DLT and Blockchain - What is a DLT?

Distributed Ledger Technology → enables the creation of a database, shared and synchronised among multiple nodes of a geographically distributed network.
Smart Contracts, Nodes and the Consensus mechanism are fondamental components of the system.

SMART CONTRACTS

self-executing programmable agreement that runs on top of a decentralized ledger

→ Enabling automated, trustless transactions/processes without intermediaries

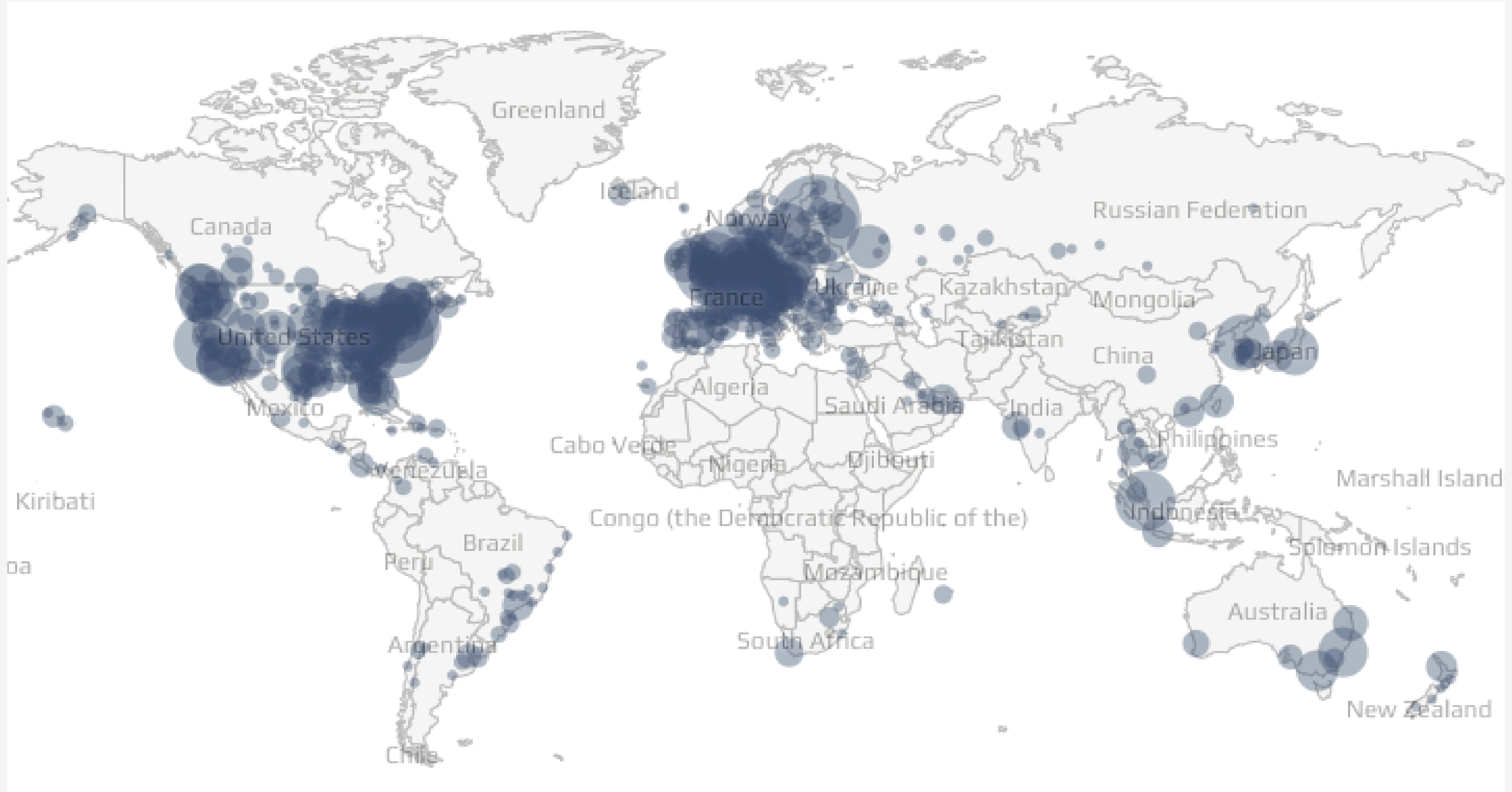
→ Executed by

NODES

Geographically distributed

→ Decentralization

→ Validators:
- confirm data validity;
- upload data through a Consensus Mechanisms



DLT and Blockchain - Consensus Mechanisms

Is a set of rules and processes used to agree on the validity of data, transactions and the state of the ledger by the participants of the network.
Crucial aspect for achieving agreement in a decentralized system where there is no central authority.

Proof of Work (POW)	Bitcoin
Proof of Stake (POS)	Ethereum
Proof of Authority (POA)	VeChain
Hybrid Methods: POS + POH	Solana

They differ in the way the consensus is reached, affecting energy consumption, security, decentralization, efficiency, scalability and speed.

DLT and Blockchain - What is a Blockchain?

In a DLT, data are saved in different ways, accordingly to the structure chosen.
BLOCKCHAIN → most famous structure.
Data → organised in a subsequent chain of blocks, each one linked to the previous in a univocal way, that contain data and transactions validated periodically.

DLT and Blockchain - Hash

Each block is uniquely identified by a **cryptographic hash**.
Cryptographic hash → entirely dependent on the block's content, it contains the parent hash.

Blocks permanently locked

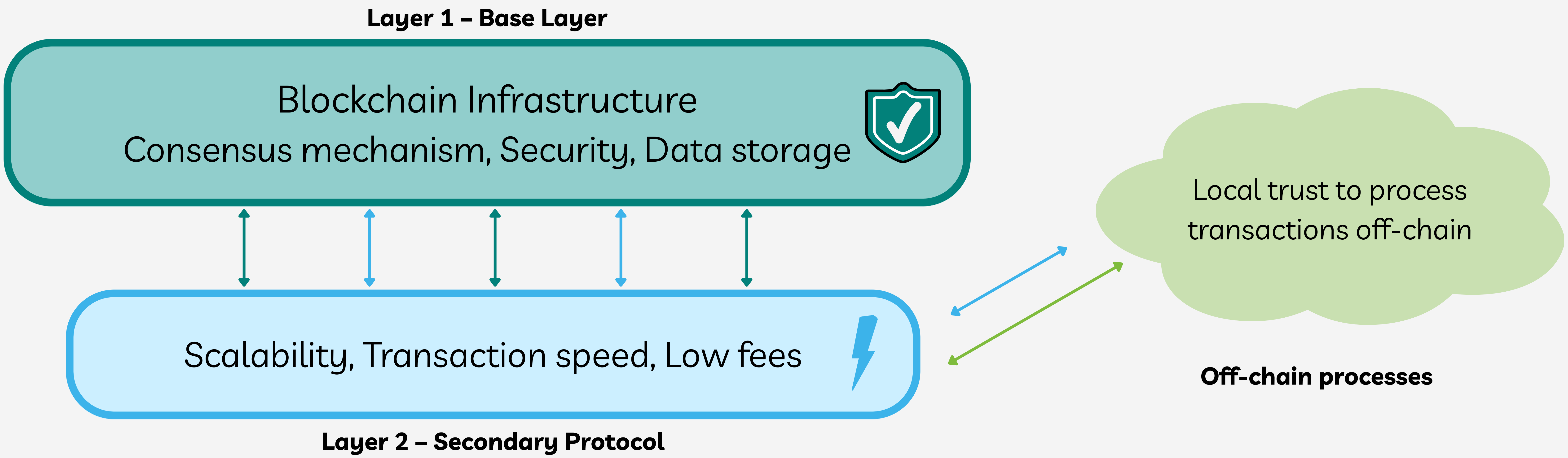
Integrity
and
immutability
for the whole blockchain

DLT and Blockchain - Different Layers

Bitcoin L1 and Lightning Network L2
Ethereum L1 and Polygon L2.

Blockchains can have more than one layer.

Layer 1 → creates the main framework, the infrastructure
Layer 2 → a protocol, built on top of L1 and interacting with it, to enhances characteristics.
Some L2 use "local trust" to process transactions almost instantaneously off-chain before sending them to main net.



DLT and Blockchain - Wallets

Wallet → generated by a private key (seed).

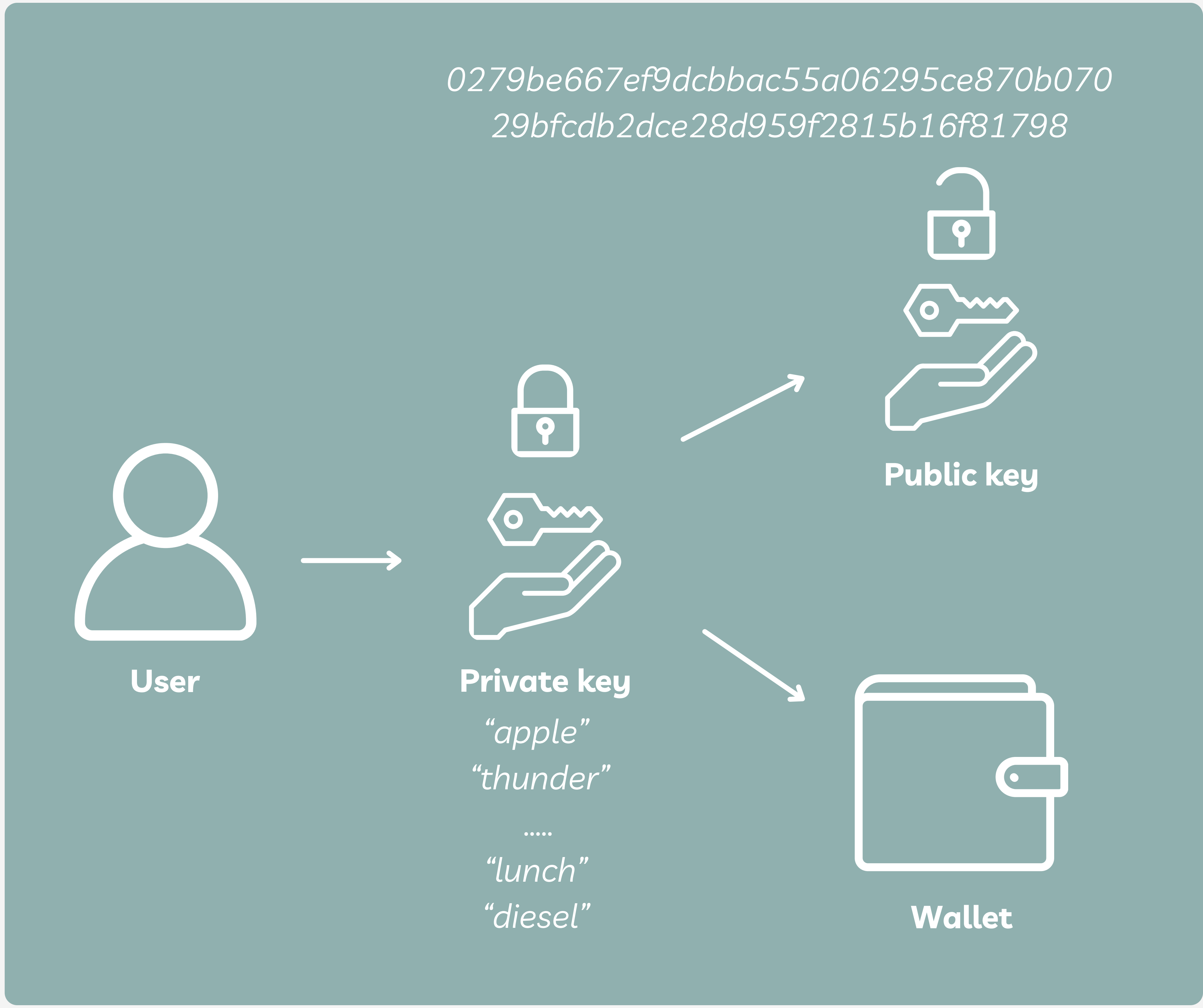
Private key

↓

unique cryptographic code

↓ used to

digitally sign transactions
recover the access to the wallet



DLT and Blockchain - Wallets

Based on the actor who safely stores the private key, it is possible to identify two different types of wallets.

Custodial (Trusted) Wallets	Non Custodial (Trustless) Wallets
• Similar to bank accounts • Generated and menaged by third parties • In case of loss of passphrase → the Central Authority can recover it • if Central Authority gets hacked → private keys are compromised and funds inside wallets lost	• Users have complete control • Users → responsible for securing their private keys and executing transactions • In case of loss → the recovery is no possible • Maximal Security

DLT and Blockchain - Permissioned vs Permissionless

Classification based on access control:

PERMISSIONLESS (public)	PERMISSIONED (private)
• Open networks ⇒ joining, partecipation and interactions without any special permission • Complete decentralization ⇒ no control by a single entity • Transparent encrypted transactions visible to anyone.	• Closed network ⇒ Central enthity controls the access and menages the ecosystem • Require user identification and role assignment • Partially decentralised / fully centralised • Enhanced security features
Examples: Bitcoin and Ethereum, Polygon	Example: Quorum by JPMorgan, Polygon

DLT and Blockchain - Existing blockchain

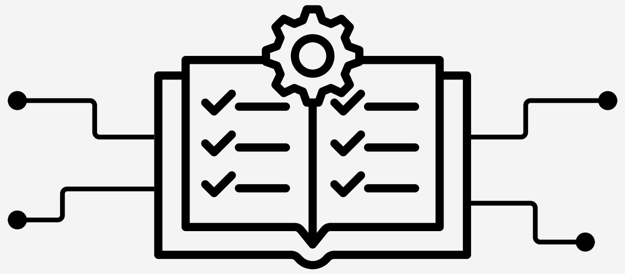
Blockchain	Consensus Mechanism	Latency [sec]	Transaction per second (TPS)	Number of nodes	Cost [$]
Bitcoin Layer 1	Proof of Work	600	7	20.500	5
Lightning Network BTC L2	Proof of Work	0,001 - 1	1 million	20.500	0,00003
Ethereum L1	Proof of Stake	12	119	1.070.000	1,30 (peaks: 50-80)
Polygon L2	Proof of Stake	2,13	7.200	100	0,01 (peaks: 0,10)
Solana L1	Proof of Stake + Proof of History	0,4	65.000	4.000	0.00025

Interoperability

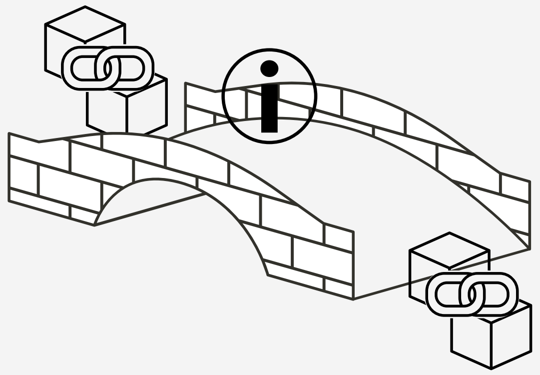
Ability of two or more blockchain systems to operate together, despite differences in their networks, consensus mechanisms and smart contract languages.

Three elements needed:

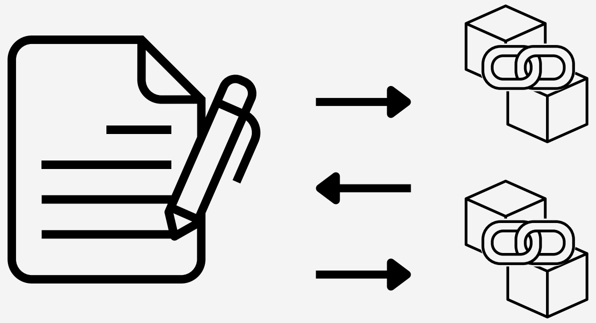
INTEROPERABILITY PROTOCOLS



CROSS CHAIN BRIDGES



SMART CONTRACTS "CROSS-CHAIN AWARE"



Interoperability

Multi-Chain Platform	Cross-Chain Network
→ Decentralized infrastructure connecting different blockchains → Enable communication in a single ecosystem	→ Connect different blockhain through an orchesration layer → No direct coding for each chain
+ Control + Security High upfront cost	+ Scalability + Transaction speed High transaction cost
Cosmos (ATOM), Polkadot and Chainlink	Axelar and LayerZero

Case Study - Cassa Depositi e Prestiti

First digital bond issuance in Italy exploiting DLT technology.

18/07/2024

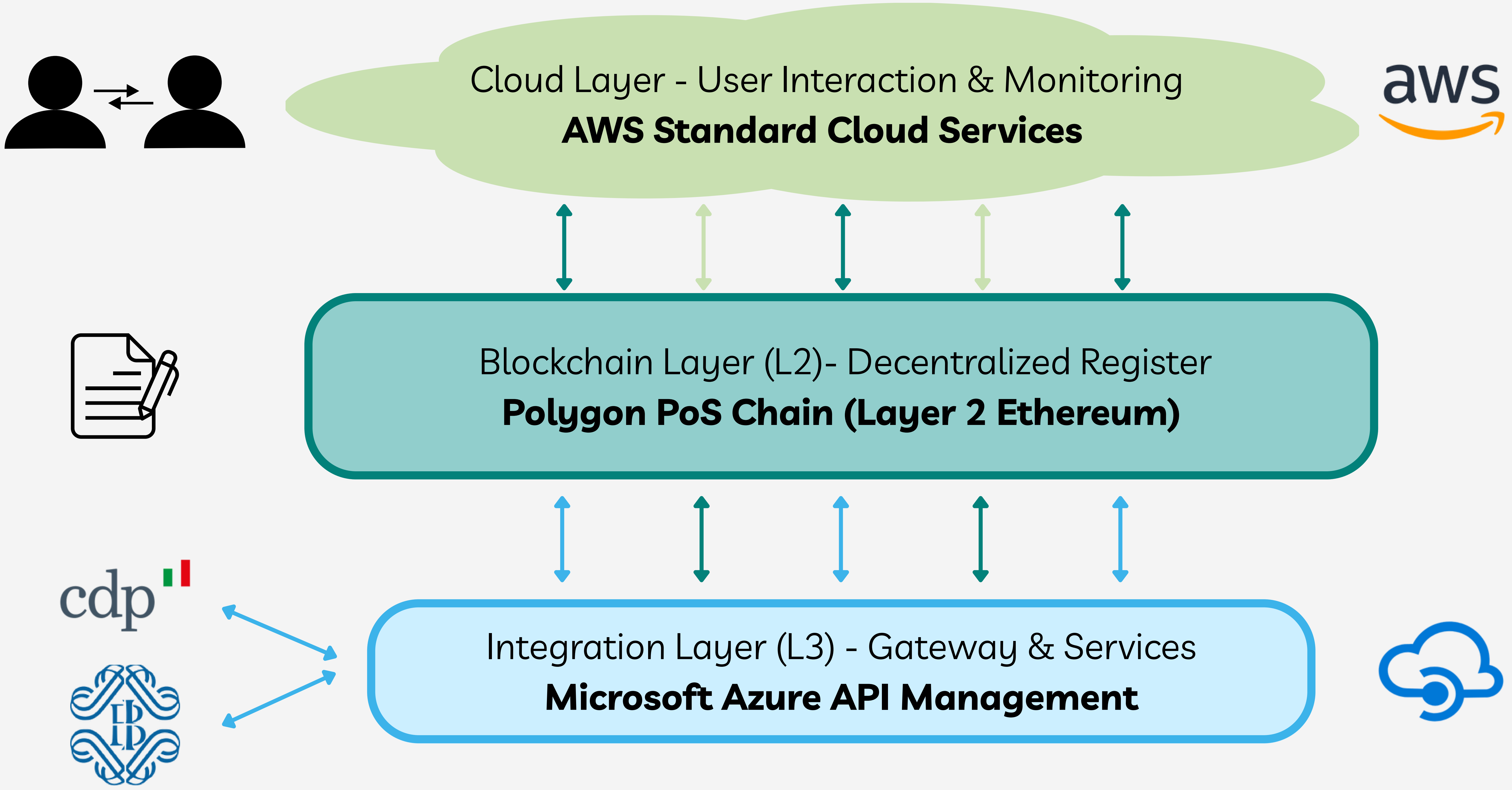
For Institutional Investors only

Digital bond for a value of €25M

DLT technology and Polygon PoS Blockchain

Smart Contract control + on-chain and off-chain functions

Case Study - Cassa Depositi e Prestiti



Digital currencies - CBDC

Central Bank Digital Currencies (CBDC) are digital currencies issued and backed by national central banks.

Trust
Stability → faster, more secure, and programmable digital payments

Attributes:

- grade of decentralization
- programmability of transactions
- privacy and data governance mechanisms
- regulation

Digital Pound
e-CNY

Digital currencies - Stablecoins

Digital tokens issued by private entities and typically anchored to fiat currencies, most commonly at a 1:1 ratio.

Built to operate on public blockchains

Already serving a wide range of use cases in:

- decentralized finance (DeFi)
- cross-border payments
- tokenized asset markets

USDC
EURC

Digital currencies - Digital Euro

The most relevant and promising CBDC, issued by European Central Bank and planned to be launched around 2028.

Designed to complement physical cash while preserving the Eurozone's integrity and monetary stability

Key features relevant for the analysis:

- Level of programmability and support of automated transactions
- Level of decentralization of chosen infrastructure

Distribution will happen through **supervised intermediaries** which will take care of creation and management of digital wallets

Banks

Payment Service Providers

3

Regulatory Landscape

Key Regulations

CORE REGULATIONS

This supranational legislatory set makes the core part of application of the analysis

DLT Pilot Regime	**Financial Market Infrastructure (FMI)** Sets the rules for MTF, TSS and SSs to issue, trade and settle financial instruments on a Distributed Ledger Technology, with some limitations: • Ceiling per bond: < € 1 Billion • Ceiling for total instruments: < € 6 Billion at the start, < € 9 Billion with transition strategy.
MiFID II	**Underlying Securities** • Defines common rules to regulate financial markets • Data Reporting Services Provider and are specified powers on the hand of competent authorities with the collaboration of ESMA.

Key Regulations

CSDR	**Depositary, Clearing, Settlement** Common rules defined for all CSDs to operate; prevent, monitor and manage settlement failures; LEI code that identifies every single client
MICAR	**Crypto Assets and Tokenized Assets** • A legal person/credit institution can receive the authorisation to offer or seek the admission to trade asset-referenced tokens but must be compliant with norms defined in the regulation, such as transparency and the constant publication of information regarding the crypto asset. • Own funds required at any time
AML & KYC	**Anti-money laundering & Customer Identification -> Wallet Creation** • License required if Euronext becomes portfolio issuer • The issuing procedure will be like opening any account
Prospectus Regulation	**Digital bonds issuing disclosure rules for Prospectus publication**

4 Solutions

4.1 Market Readiness

4.2 Advantages of DLT and Blockchain

4.2 Solution 1

- **4.2.1** Case A
- **4.2.2** Case B
- **4.2.3** Case C
- **4.2.4** Case D
- **4.2.5** Case E

4.4 Solution 2

4.5 Market Impact

Market Readiness

Regulatory readiness

Today's market → set of standing regulations for a unique European financial market.

Directives must be implemented as primary national legislation.

There is a strong signal of interest from market players to implement a more efficient, faster and distributed ledger technology

First movers

May 2024 → Euroclear and Clearstream have released, together with BCG and DTT, a framework for the adoption of a digital assets market, including that of tokenized assets.

Advantages of DLT and Blockchain

	Current Infrastructure	Potential DLTs
Settlement time	T+2	Potential T+0
Costs	Human and technological capital intensive → High CAPEX and OPEX	Group's infrastructure simplified → low thanks to smart contracts
Scalability	Optimized for large volumes but require investments	High, investments needed to optimize latency and energy consumption
Transparency	Limited, access granted only to authorized players	High, multiple parties can access transaction data

Advantages of DLT and Blockchain

	Current Infrastructure	Potential DLTs
Intercompatibility	Medium-Low due to presence of multiple CSDs across countries; limited readiness for blockchain-based trading	High, DLTs support innovative programmable features
Standardization	Medium-Low due to presence of multiple CSDs across countries; limited readiness for blockchain-based trading	High, DLTs support innovative programmable features
Reliability	High but exposed to force majeure events (e.g. localized blackouts, hacking...)	Enhanced resiliance thanks to decentralization and cryptography
Riskiness	High: presence of Margin&Default Fund equal to €25.1 B as collateral	Better management of default risk and related risks

Proposed Solutions

Use of Blockchain → useful mostly for CDS and Clearing&Settlement areas.

Benefits

- All parties can access the same shared ledger in real time
- No need for constant data checks between systems
- Faster and more secure transactions
- Long term → settlement and clearing in T+0
- Increased trust and reduced errors and scams

Six parameters used for the evaluation of each solution:

- ✓ Decentralization
- ✓ Transaction speed
- ✓ Scalability
- ✓ Safety - Cybersecurity and Energy security
- ✓ Transaction cost - compared with current infrastructure
- ✓ Intercompatibility between different systems

Proposed Solutions

Different solutions have been identified

Four key actors: Euronext, Regulators/Financial Authorities, Issuers and Investors.

EURONEXT	Central authority and network regulator Validator and gatekeeper Network manager Ledger maintenance
REGULATORS AND FINANCIAL AUTHORITIES	Granted with observer and validator access Real time transactions monitoring Regulatory compliance verification Interventions in case of anomalies or sistemic risks
ISSUERS & INVESTORS	Configuration vary across different solutions

Solution 1

Implementation of a Permissioned Blockchain based on the adoption of existing infrastructures as Ethereum, Polygon L2 and Solana.

PERMISSIONED → to ensure access exclusively to authorised users in compliance with the corresponding profiles.

EURONEXT → storage of encrypted data on a public worldwide blockchain.

✓ Persistency ✓ No geographical risk ✓ Low fees ✓ High speed

Solution 1.A - Short term

SOLUTION FOR RETAIL INVESTORS

- Blockchain integrated only in Clearing&Settlement activities
- Semi-automated smart contract → still need payment's input
- Confirmations by the intermediary required

MOT	**Retail**
	Institutional & Professional
EURONEXT ACCESS	**Institutional & Professional**

	Decentralization	Safety	Transaction speed	Transaction costs	Scalability	Intercompatibility
LOW						
MEDIUM						
HIGH						

Solution 1.A - Short term

MOT	Retail
	Institutional & Professional
XT	Institutional & Professional

SOLUTION FOR RETAIL INVESTORS

- Blockchain integrated
- Semi-automated Sr
- Confirmations by th

Decentralization doesn’t change in respect to the current situation.

	Decentralization	afety	speed	costs	Scalability	Intercompatibility
LOW						
MEDIUM						
HIGH						

Solution 1.A - Short term

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR RETAIL INVESTORS

- Blockchain integrated only in Clearing&Settlement activities
- Semi-automated Smart contract → still need payment's input
- Confirmations by the intermediary

Slightly higher thanks to the smart contract that reduces data checks and errors but there are no transaction to be recorded on the blockchain.

	Decentralization	Safety				...tibility
LOW						
MEDIUM						
HIGH						

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Solution 1.A - Short term

MOT	Retail
	Institutional & Professional
EURONEXT	...nal

SOLUTION FOR RETAIL INVESTORS

- Blockchain integrated only in Clearing&Settlem...
- Semi-automated Smart contract → still need...
- Confirmations by the intermediary required

Transaction Speed = T+2
It doesn't change in respect to the current situation.

	Decentralization	Safety	Transaction speed	... costs		
LOW						
MEDIUM						
HIGH						

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Solution 1.A - Short term

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR RETAIL INVESTORS

- Blockchain integrated only in Clearing&Settlement activities
- Semi-automated Smart contract → still need payment's input
- Confirmations by the intermediary required

There is very low benefit from the usage of blockchain. Reduction in errors and double check matching.

				Transaction costs	Scalability	Intercompatibility
LO						
MEDIUM						
HIGH						

Solution 1.A - Short term

SOLUTION FOR RETAIL INVESTORS

- Blockchai
- Semi-au
- Confirm

The limitation is given by semi-automatic smart contracts: only a small percentage of the blockchain potential is exploited.
No incentive for external players to join the network.

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

	De				Scalability	Intercompatibility
LOW						
MEDIUM						
HIGH						

Solution 1.A - Short term

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	Retail
MOT	Institutional & Professional
	Institutional & Professional

SOLUTION FOR RETAIL INVESTORS

- Blockchain integrated only in C
- Semi-automated Smart cont
- Confirmations by the interm

The system is totally closed and any interconnection with other players and systems would be time and money intensive.

	Decentralization	Safety				Intercompatibility
LOW						
MEDIUM						
HIGH						

Solution 1.B - Short Term

MOT	Retail
MOT	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR INSTITUTIONAL AND PROFESSIONAL INVESTORS

- Banks and Issuers → custodial wallet + private key stored by Euronext
- Smart contracts → allow transfer of tokenized bonds between wallets
- Transactions → recorded in the blockchain

	Decentralization	Safety	Transaction speed	Transaction costs	Scalability	Intercompatibility
LOW				■		■
MEDIUM	■		■		■	
HIGH		■				

Solution 1.B - Short Term

SOLUTION FOR INSTITUTIONAL AND PROFESSIONAL INVESTORS

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

- Banks and Issuers → custodial wallet + private key stored by Euronext
- Smart Contracts → allow
- Transactions → recor

The system is more decentralized thanks to wallets. Since Euronext stores the private key of every wallets it may become a target for hackers.

	Decentralization					Intercompatibility
LOW						
MEDIUM						
HIGH						

Solution 1.B - Short Term

MOT	Retail
MOT	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR INSTITUTIONAL AND PROFESSIONAL INVESTORS

- Banks and Issuers → custodial wallet + private key stored by Euronext
- Smart Contracts → allow transfer of tokenized bonds between wallets.
- Transactions → recorded in the blockchain

	Decentralization	Safety				Intercompatibility
LOW						
MEDIUM						
HIGH						

Transactions are recorded in the blockchain enhancing the current situation.

Solution 1.B - Short Term

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR INSTITUTIONAL AND PROFESSIONAL INVESTORS

- Banks and Issuers → custodial wallet + private key stored by Euronext
- Smart Contracts → allow transfer of tokenized bonds between wallets
- Transactions → recorded in the blockchain

	Decentralization	Safety	Transaction speed			ity
LOW						
MEDIUM						
HIGH						

The transaction speed of the digital bond is almost instantaneous but the payment confirmation is a limitation.

Solution 1.B - Short Term

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR INSTITUTIONAL AND PROFESSIONAL INVESTORS

- Banks and Issuers → custodial wallet + private key stored by Euronext
- Smart Contracts → allow transfer of tokenized bonds between wallets.
- Transactions → recorded in the blockchain

	Decentralization	Safety	Transaction speed	Transaction costs	Scalability	Intercompatibility
LOW						
MEDIUM						
HIGH						

Physical transfer of tokenized bond through blockchain transaction.

Solution 1.B - Short Term

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR INSTITUTIONAL AND PROFESSIONAL INVESTORS

- Banks and Issuers → custodial wallet + private key stored by Euronext
- Smart Contracts → allow transfer of tokenized bonds between wallets.
- Transacti

The marginal costs to create new wallets are near zero.
No incentive for external players to join the network.

					Scalability	Intercompatibility
LOW						
MEDIUM						
HIGH						

Solution 1.B - Short Term

SOLUTION FOR INSTITUTIONAL AND PROFESSIONAL INVESTORS

- Banks and I…
  Euronext
- Smart Con…
- Transactio…

	MOT	
		Retail
		Institutional & Professional
		…stitutional & Professional

Users can only use their wallets in this specific network.
Direct connection between Euronext and the buyers through the tokenized bonds
⇒ the compatibility is limited within Euronext closed system.

	Decent…					Intercompatibility
LOW						
MEDIUM						
HIGH						

Solution 1.C - Long Term

SOLUTION FOR RETAIL INVESTORS

- Euronext → creates a wallets for each retail banks
- Banks → generate individual blockchain wallets for customers and manages the digital bond transaction
- Smart contracts → not totally automatic → traditional payment flow

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

	Decentralization	Safety	Transaction speed	Transaction costs	Scalability	Intercompatibility
LOW						
MEDIUM						
HIGH						

Solution 1.C - Long Term

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR RETAIL INVEST

- Euronext → creates a wo
- Banks → generate individu s
  and manages the dig
- Smart Contracts → not totally automatic → traditional payment flow

All parameters are evaluated equal to the previous Solution.

	Decentralization	Safety	Transaction speed	Transaction costs	Scalability	Intercompatibility
LOW						
MEDIUM						
HIGH						

Solution 1.D - Long Term

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR ALL INVESTORS WITH NON-CUSTODIAL WALLETS
FOR BANKS AND ISSUERS

- Euronext → authorises functions and permits
- Cyberattaks → don't affect the whole system → compartmentalization

	Decentralization	Safety	Transaction speed	Transaction costs	Scalability	Intercompatibility
LOW						
MEDIUM						
HIGH						

Solution 1.D - Long Term

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR ALL INVESTORS WITH NON-CUSTODIAL WALLETS FOR BANKS AND ISSUERS

- Euronext → authorises functions and permits
- Cyberattaks → don't affect the whole system → compartmentalization

	Decentralization	Safety	Transaction	Transaction	Scalability	Intercompatibility
LOW						
MEDIUM						
HIGH						

Issuers, Banks, Institutional and Professional investors have a non-custodial wallet.

Solution 1.D - Long Term

SOLUTION FOR ALL INVESTORS WITH NON-CUSTODIAL WALLETS FOR BANKS AND ISSUERS

- Euronext → authorises functions and permits
- Cyberattaks → don’t affect the whole system → compartmentalization

MOT	Retail	
	Institutional & Professional	
EURONEXT ACCESS	Institutional & Professional	

	Decentralization	Safety				compatibility
LOW						
MEDIUM						
HIGH						

The system is compartimentalized.
The probability of cyberattacks increase with the spread of wallets, but the impact is lower than an attack on the entire centralized system.
Moreover there is a mismatch of wallets between money flow (bank wallets) and tokens flow (blockchain wallets).

Solution 1.D - Long Term

SOLUTION FOR ALL INVESTORS WITH NON-CUSTODIAL WALLETS FOR BANKS AND ISSUERS

- Euronext → authorises functions and permits
- Cyberattaks → don't affect the whole system → compartmentalization

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

	Decentralization	Safety	Transaction speed			ility
LOW						
MEDIUM						
HIGH						

The transaction speed of the digital bond is almost instantaneous but the payment confirmation is a limitation.

Solution 1.D - Long Term

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR ALL INVESTORS WITH NON-CUSTODIAL WALLETS FOR BANKS AND ISSUERS

- Euronext → authorises functions and permits
- Cyberattaks → don't affect the whole system → compartmentalization

	Decentralization	Safety	Transaction speed	Transaction costs	Scalability	Intercompatibility
LOW						
MEDIUM						
HIGH						

Physical transfer of tokenized bond through blockchain transaction.

Solution 1.D - Long Term

MOT		Retail
		Institutional & Professional
EURONEXT ACCESS		Institutional & Professional

SOLUTION FOR ALL INVESTORS WITH NON-CUSTODIAL WALLETS FOR BANKS AND ISSUERS

- Euronext → authorises functions and permits
- Cyberattaks → don't affect the whole system → compartmentalization

	Decentralizati		Transaction	Transaction	Scalability	Intercompatibility
LOW						
MEDIUM						
HIGH						

Banks and intermediaries are more willing to adopt this solution since they would fully manage the wallets of their clients.

Solution 1.D - Long Term

SOLUTION FOR ALL INVESTORS WITH NON-CUSTODIAL WALLETS FOR BANKS AND ISSUERS

- Euronext → authorises functions and permits
- Cyberattaks → don’t affect the whole system → compartmentalization

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

	Decentralization				ility	Intercompatibility
LOW						
MEDIUM						
HIGH						

Institutional&Professional investors can use the same non-custodial wallet for different exchanges and blockchains but it is still missing the direct interconnection between the money flow and smart contracts.

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Solution 1.E - Long Term

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

SOLUTION FOR ALL INVESTORS WITH DIGITAL EURO INTEGRATION

Starting point: solution C and D

Integration of Digital Euro with digital wallets → full exploitation of smart contracts potential.

Payment transactions → almost instantaneous.

>>> Usage of multiple digital currencies and stablecoins (Digital Pound, e-CNY, USDC) >>> Expansion into new markets

! The successful deployment critically depend on the ability to ensure both technological interoperability and strict compliance with the regulatory frameworks specific to each currency involved. !

Solution 1.E - Long Term

		Safety	Transaction speed	Transaction costs	Scalability	Intercompatibility
MEDIUM						
HIGH			T+0			

The consequences of a private key being stolen are more impactful since tokens and digital currencies could be stored in the same wallet.

There is also money flow inside the system

Complete intercompatibility between different blockchains, wallets and payment methods.

Solution 2

BUILDING A PRIVATE BLOCKCHAIN FROM SCRATCH

- Private Permissioned Blockchain → internal servers as nodes
- Higher initial investment

	Decentralization	Safety	Transaction speed	Transaction costs	Scalability	Intercompatibility
LOW	System fully centralized within Euronext	Euronext servers				
MEDIUM			Euronext servers			
HIGH						Depending on the set up

Market Impact - Collapsing positions

* referred to solution 1.E

wallets for all investors → External infrastructures (SWIFT, SEPA and intermediated payment systems) / Clearing & Settlement → SIGNIFICANTLY STREAMLINED

Digital Euro in the long term scenario → Three main strategic impacts:

(A) Operational Transformation
Transaction time decrease → matching, clearing, and settlement within the same day

T+2 → T(0)

→

(B) Capital&Liquidity Efficiency
Reduced collateral requirements at CCPs → release of liquidity and securities deposited in the market

€ 25B released

→

(C) Change in strategic position
From a volume-based to a subscription-based business + one-time services model

Drift from volume-based business

POLITECNICO MILANO 1863

Market Impact - Strategic positioning

BASE ASSUMPTIONS:

- CSD revenue growth to €270,5M (+8,7% in 2024) - thanks to increased settlement flows and growth in assets deposited as collateral growth;
- Higher speed of transaction ⇒ a) Slightly lower fees (to attract trades);
  b) Differentiation based on T(0) and ≃ 0 deposit;
  c) Increase in custodial service fees.
- Medium to long-term impact of this technology within the market → underestimated.

The Strategic Shift & Main Insights

First-mover: T+0 infrastructure integrated with a Digital Euro ⇒ CCP and custodian services run the risk of becoming commoditized thanks to greater decentralization and lower regulatory pressure.

- Implementation of a blockchain infrastructure ⇒ strategic shift in Euronext's business model: **from a volume-based business towards a subscription-based and one-time services model**.
- Main value that Euronext offers is access to capital markets, checking legal requirements and data analyses. The solution **strengthen the fixed income business area and develop the capital market access area**.

5 Sustainability & Environmental Impact

5.1 Energy Consumption Overview

5.2 Cases Study

- 5.2.1 Cassa Depositi e Prestiti
- 5.2.2 Société Générale FORGE

5.3 Challenges and Strategic Considerations

Energy Consumption Overview

How to assess correctly the DLT and Blockchain's energy impact ?

Three aspects to consider: → Traditional Alternative
→ Value Created
→ Technological Features

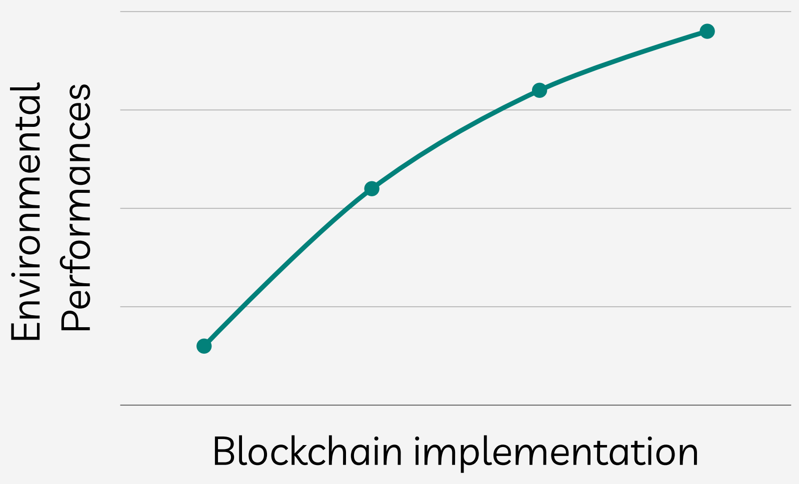
Traditional Alternative

Blockchain Technology vs Financial Sector → Direct comparison not appropriate

Financial Systems → High energy footprint

Traditional banking system:
260TWh/year

Energy Consumption Overview - Value Created



Thanks to traking, transparancy and automation capabilities

Improved Data Collection
Leveraging the transparent and auditable nature of blockchain.
Record of ESG data → more informed investments decision.

Tamper-proof real-time tracking
Accurate record of GHG emissions and compliance information → better control and risk mitigation.

Tokenized Green Bond
Increased transparency → Investors can track track the positive environmental impact of projects in near real-time.

Energy Consumption Overview - Tech Features

Different Technological features ⇒ different environmental impacts

CONSENSUS MECHANISM	Relevant in terms of carbon and environmental footprint
SCALABILITY	Low scalability ⇒ High energy usage
LATENCY (CONFIRMATION TIME)	High latency ⇒ inefficient consensus mechanism ⇒ High energy usage
THROUGHPUT (TRANSACTIONS PER SECOND)	High throughput ⇒ less blocks needed per transaction ⇒ Low enegergy usage
DESIGN OF NODES DISTRIBUTION	High number of nodes ⇒ redundancy ⇒ high energy usage
INTEROPERABILITY	Minor increment in energy footprint due to cross-network transactions

Energy Consumption Overview - Tech Features

Consensus Mechanism	Proof of Work (PoW) BTC L1	Proof of Stake (PoS)	Delegated Proof of Stake (DPoS)	Proof of Authority (PoA)	Hybrid methos
Energy per transaction [KWh]	707,6	0,002	0,0015	0,00022	2,4
CO2 emission per transaction [g]	380.000	0,8	0,05	0,03	45
Scalability	Low	High	Medium	Medium	Medium
Latency [sec]	600	15	1	<1	60
Throughput (TPS)	7	1000	3000	2000	300
Sustainability	LOW	HIGH	HIGH	HIGH	MEDIUM

Cases Study - Cassa Depositi e Prestiti

“BlockInvest” Software

Polygon blockchain -
Layer 2 of Ethereum

Technology’s Impact

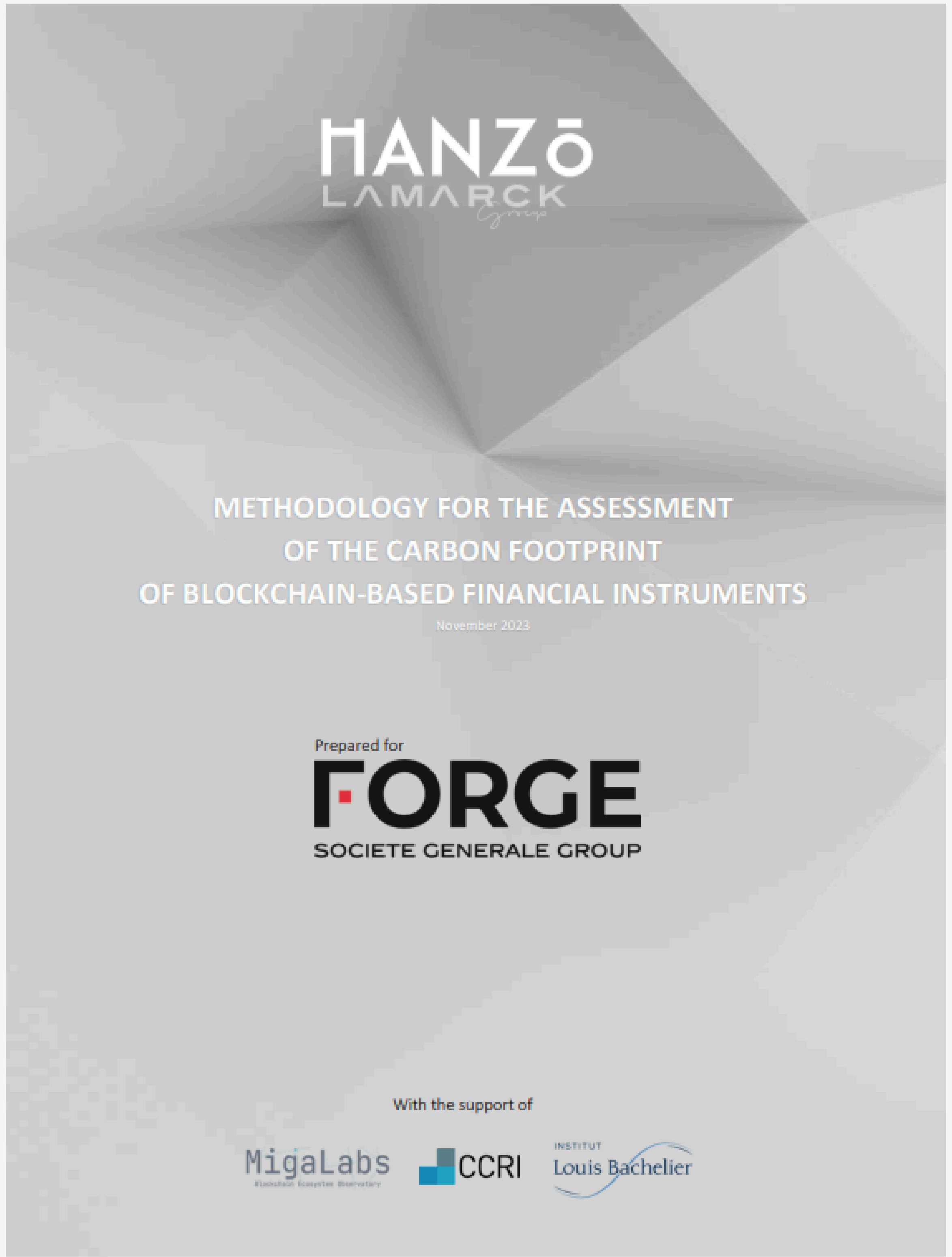
PoS Consensus mechanism

- low energy per transaction
- high scalability

Community Engagement

Polygon support eco-friendly initiatives within the blockchain space.

Cases Study - Société Générale FORGE



Ethereum Blockchain

« SG-Forge believes that blockchain technology has the potential to play an important role in addressing climate change [...]. SG-Forge also believes that blockchain-based market infrastructures can have a lower carbon footprint than current infrastructures. »

Focus of the report: environmental impact of the bond issuance after “*The Merge*”

Ethereum: Proof of Work → Proof of Stake

ΔEnergy = - 99,98%

Challenges and Strategic Considerations

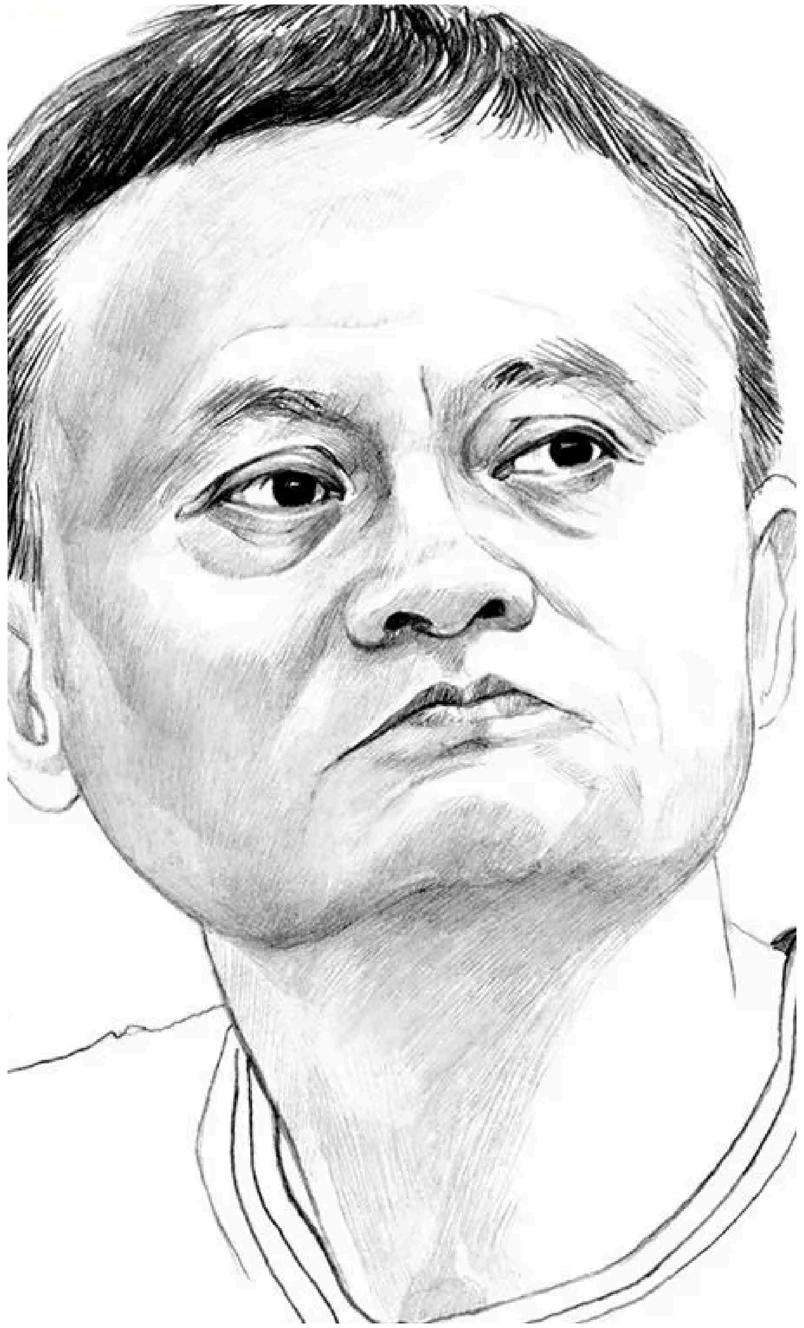
Strategic choice of the right consensus mechanism

Optimization of the network design
(latency, scalability, TPS…)

Possible value created on the
ESG reporting requirements

6 Conclusions

Conclusion



"Blockchain will fundamentally change the financial systems in the next 10 to 15 years. A blockchain technology will be applied in many areas because it is about trust, credit, security — the security of data and the privacy of data."

— Jack Ma, Forbes Blockchain 50, 2021

Thanks for your attention!

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